

EXPERIMENT NO:4

Topic: Automatic Door Lock system.

Aim:

To design and implement a secure door lock system using RFID authentication with Arduino UNO. This project enhances security by implementing contactless access control using RFID tags and an electromagnetic lock.

Introduction:

With the advancement of technology, traditional lock-and-key mechanisms are being replaced by smart authentication systems. An RFID-based door lock system provides a modern and efficient solution for access control. Radio Frequency Identification (RFID) technology enables contactless authentication, making it more convenient and secure.

This project employs an RFID reader to scan RFID tags or cards assigned to authorized users. When a valid tag is detected, the system triggers an electromagnetic lock to grant access. The entire process is managed using an Arduino UNO, which processes RFID data and controls the locking mechanism. Unlike traditional password-based systems, this method eliminates the need for remembering or inputting codes, reducing the risk of unauthorized access due to weak passwords.

Working Principle:

1. The RFID reader scans the RFID card/tag.
2. The unique ID of the tag is sent to the Arduino UNO.
3. The Arduino compares the scanned ID with the stored authorized IDs in its memory.
4. If the ID matches, the Arduino sends a signal to unlock the door.
5. If the ID is unauthorized, access is denied, and an alert (such as an LCD) may be triggered

Theory of Components Used in RFID-Based Door Lock System

1. Arduino UNO

A microcontroller board that controls the entire system. It reads RFID data, verifies access, and operates the servo motor to lock/unlock the door.

2. RFID Module (RC522)

An RFID reader that operates at 13.56 MHz, used to scan RFID tags/cards and send the unique ID (UID) to the Arduino for authentication.

3. RFID Tag/Card

A contactless tag with a unique identification number. When placed near the RFID reader, it transmits its ID, which is checked for authorization.

4. Servo Motor

A servo motor is used as a locking mechanism. Upon successful authentication, the Arduino rotates the servo to unlock the door. After a few seconds, it returns to its original position to relock the door.

5. LCD Display (16x2)

An LCD screen is used to provide real-time feedback to users. It displays messages such as:

- "Scan Your Card" when waiting for input.
- "Access Granted" if an authorized RFID tag is scanned.
- "Access Denied" for unauthorized tags.

INTRODUCTION TO ARDIUNO UNO

The Arduino Uno is a microcontroller board based on the ATmega328P, widely used in embedded systems, automation, and IoT applications due to its ease of use, open-source nature, and extensive community support. It operates at 5V with a 16 MHz clock speed and features 14 digital I/O pins, 6 analog inputs, and communication interfaces like UART, SPI, and I2C. The board is programmed using the Arduino IDE, making it ideal for beginners and professionals. Its versatility allows it to interface with various sensors, actuators, and displays, making it a popular choice for home automation, robotics, and security systems. In this project, the Arduino Uno acts as the central controller, processing RFID authentication, controlling the servo motor for locking/unlocking, and displaying real-time access status on an LCD screen.

CODE:

RFID tag scan code

* <https://srituhobby.com>

*/

```
#include <LiquidCrystal_I2C.h>
```

```
#include <SPI.h>
```

```
#include <MFRC522.h>
```

```
#define RST_PIN 9
```

```
#define SS_PIN 10
```

```
byte readCard[4];
```

```
byte a = 0;
```

```
LiquidCrystal_I2C lcd(0x27, 16, 2); MFRC522
```

```
mfrc522(SS_PIN, RST_PIN);
```

```
void setup()
```

```
{ Serial.begin(9600);
```

```
lcd.init();
```

```
lcd.backlight(); while
```

```
(!Serial); SPI.begin();
```

```
mfrc522.PCD_Init();
```

```
delay(4);
```

```
mfrc522.PCD_DumpVersionToSerial();
```

```
lcd.setCursor(2, 0);
```

```
lcd.print("Put your card");
```

```
}
```

```
void loop() {
```

```
if ( ! mfrc522.PICC_IsNewCardPresent())
```

```
{ return 0;
```

```
}
```

```
if ( ! mfrc522.PICC_ReadCardSerial()) { return 0;
```

```
}
```

```
lcd.clear(); lcd.setCursor(0,
```

```
0); lcd.print("Scanned
```

```
UID"); a = 0;
```

```
Serial.println(F("Scanned PICC's UID:")); for
```

```
( uint8_t i = 0; i < 4; i++) { // readCard[i] =
```

```
mfrc522.uid.uidByte[i];
```

```
Serial.print(readCard[i], HEX); Serial.print("
```

```
");
```

```
lcd.setCursor(a, 1);
```

```
lcd.print(readCard[i], HEX);
```

```
lcd.print(" ");
```

```
delay(500); a
```

```
+= 3;
```

```
}
```

CONCLUSION:The RFID-based door lock system using Arduino UNO, RFID module, servo motor, and LCD display provides a secure and efficient method for access control. The system successfully reads RFID tags, verifies authentication, and unlocks the door using a servo motor. The LCD display enhances user interaction by providing real-time feedback on access status.

This project eliminates the need for traditional keys, reducing security risks associated with lost or duplicated keys. With its simple yet effective design, the system can be implemented in homes, offices, and restricted areas for improved security